← Back **De Morgan's laws**
Practice Assignment • 30 min

Due Jul 1, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Let A and B be two sets. Which one of the following represents a De Morgan's law? 1 / 1 point

- ( ) $\overline{A \cap B} = \overline{A} \cap \overline{B}$
- ( ) $\overline{A \cap B} = \overline{A} \cup B$
- (•) $\overline{A \cap B} = \overline{A} \cup \overline{B}$
- ( ) $\overline{A \cap B} = A \cup B$

✓ Nice work

The complement of the intersection of two sets is the union of their corresponding complements

2. Let A and B be two sets. Which one of the following is TRUE? 1 / 1 point

- (•) $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- ( ) $\overline{A \cup B} = \overline{A} - \overline{B}$
- ( ) $\overline{A \cup B} = \overline{A} \oplus \overline{B}$
- ( ) $\overline{A \cup B} = \overline{A} \cup \overline{B}$

✓ Nice work

De Morgan's law, the complement of the union of two sets is the intersection of their corresponding complements

3. Let A and B be two sets. Which of the following is equivalent to $\overline{\overline{A} \cap \overline{B}}$? 1 / 1 point

- ( ) $\overline{A} \cup \overline{B}$
- ( ) $A \cap B$
- ( ) $\overline{A} \cap \overline{B}$
- (•) $A \cup B$

✓ Nice work

$\overline{\overline{A} \cap \overline{B}} = \overline{\overline{A}} \cup \overline{\overline{B}} = A \cup B$